

VSC single-molecule results: CO₂ and H₂O

Weekly update

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Summary

- **Benchmark reproduced.** The CO₂ λ sweep with NEP matches the group’s Bonini-comparison benchmark at every λ , with χ both included and neglected. PySCF agrees with NEP when χ is neglected.
- **H₂O bend polaritons.** With χ neglected, the splitting is linear in λ , $\Omega_R \approx 1265 \lambda \text{ cm}^{-1}$ (125 cm^{-1} at $\lambda = 0.1$).
- **Single-molecule checks.** All 67 runs conserve energy. CO₂ is in the linear regime and its anti-crossing matches the exact linear model to 2.6 cm^{-1} .
- **Open issues.** Both molecules rotate when the cavity polarization is tilted. With χ on, H₂O’s cavity is red-detuned. The H₂O detuning scan sits $40\text{--}60 \text{ cm}^{-1}$ below simple models.

Setup

All runs use `cboamd` from `polaritonic_deep_md` (commit `f0c0ed7`). There is one cavity mode, e-mode. Both molecules are oriented so the mode of interest has its dipole along x , the component **infrared** uses: CO₂ lies along x , and H₂O has its C₂ axis along x . The timestep is 0.5 fs. Peaks are the strongest maxima of the μ_x spectrum. Frequency resolution is $33356/(\text{steps} \times 0.5) \text{ cm}^{-1}$.

Campaign	Driver	χ	Steps	Res. (cm ⁻¹)	λ	ω_c (cm ⁻¹)	Runs
CO ₂ λ sweep	NEP	on / off	10000	6.7	0–0.3 (9)	2400	17
	PySCF LDA/cc-pVDZ	off	800	83	0–0.3 (5)	2400	5
H ₂ O λ sweep	PySCF LDA/cc-pVDZ	off	800	83	0–0.3 (7)	1584	7
	PySCF LDA/cc-pVDZ	on	800	83	0.1–0.3 (5)	1584	5
CO ₂ checks	NEP	off	10000	6.7	0.1	2100–2800	20
H ₂ O checks	PySCF LDA/cc-pVDZ	off	800	83	0.3	1300–1850	13

Table 1: Run parameters. NEP models are the CO₂ models in the group repo (`tests/e2e/no-polar/mlip/models`). The bare (no-cavity) run in each sweep is $\lambda = 0$.

1. CO₂: λ sweep (benchmark reproduction)

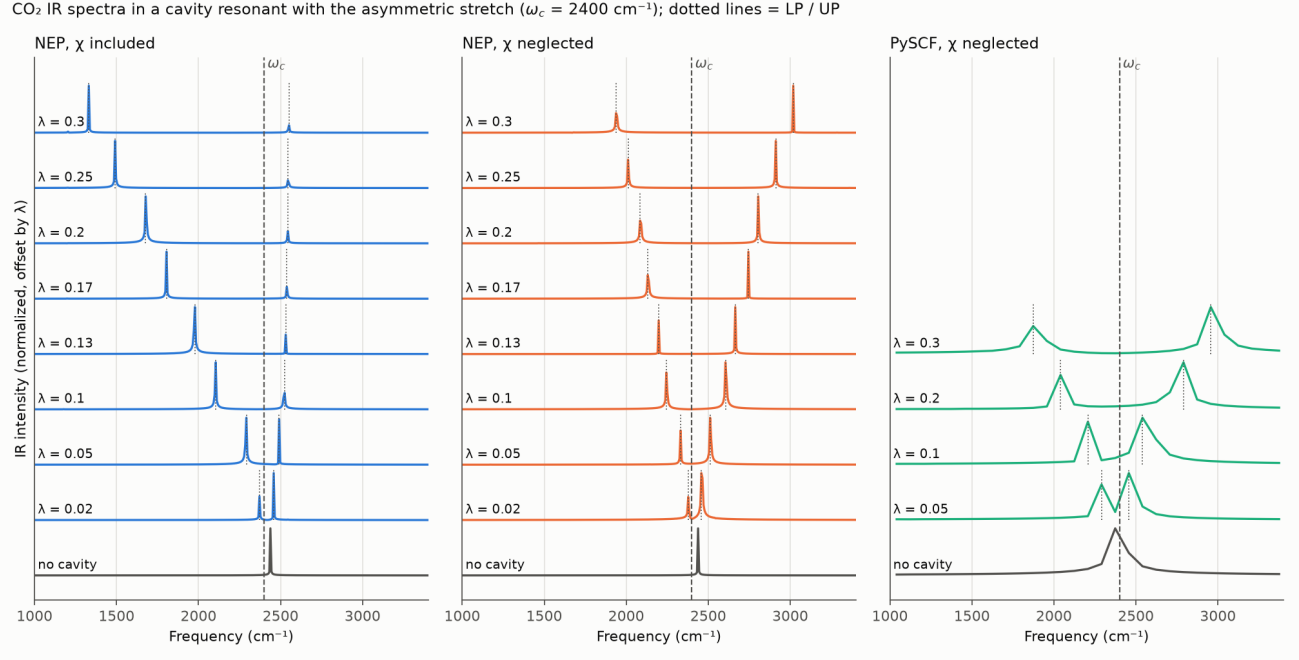


Figure 1: CO₂ IR spectra vs λ (cavity at 2400 cm^{-1}). The asymmetric stretch splits into LP and UP (dotted). With χ included, LP drops steeply and UP stays near the bare mode. With χ neglected, the splitting is roughly symmetric. PySCF peaks are broader because the runs are shorter (800 steps).

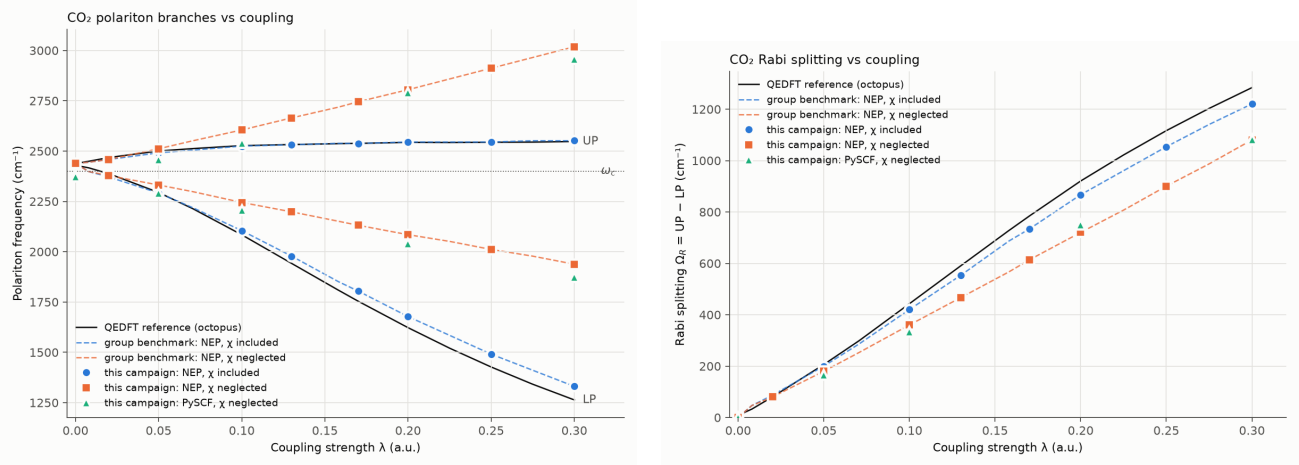


Figure 2: Left: CO₂ polariton branches. Right: Rabi splitting vs λ . Points are this work, dashed lines are the group's NEP benchmark, and the solid line is the explicit QEDFT reference (octopus).

$\Omega_R \text{ (cm}^{-1}\text{) at } \lambda =$	0.05	0.1	0.2	0.3
NEP, χ included	200	420	867	1221
NEP, χ neglected	180	360	720	1081
PySCF, χ neglected	167	333	750	1083
QEDFT reference	206	443	921	1284

Confirms: the NEP points equal the group benchmark exactly, so the setup and models are correct. PySCF matches NEP to within its 83 cm^{-1} resolution. Including χ raises Ω_R by about 15% and brings it within about 5% of QEDFT.

2. H₂O: λ sweep

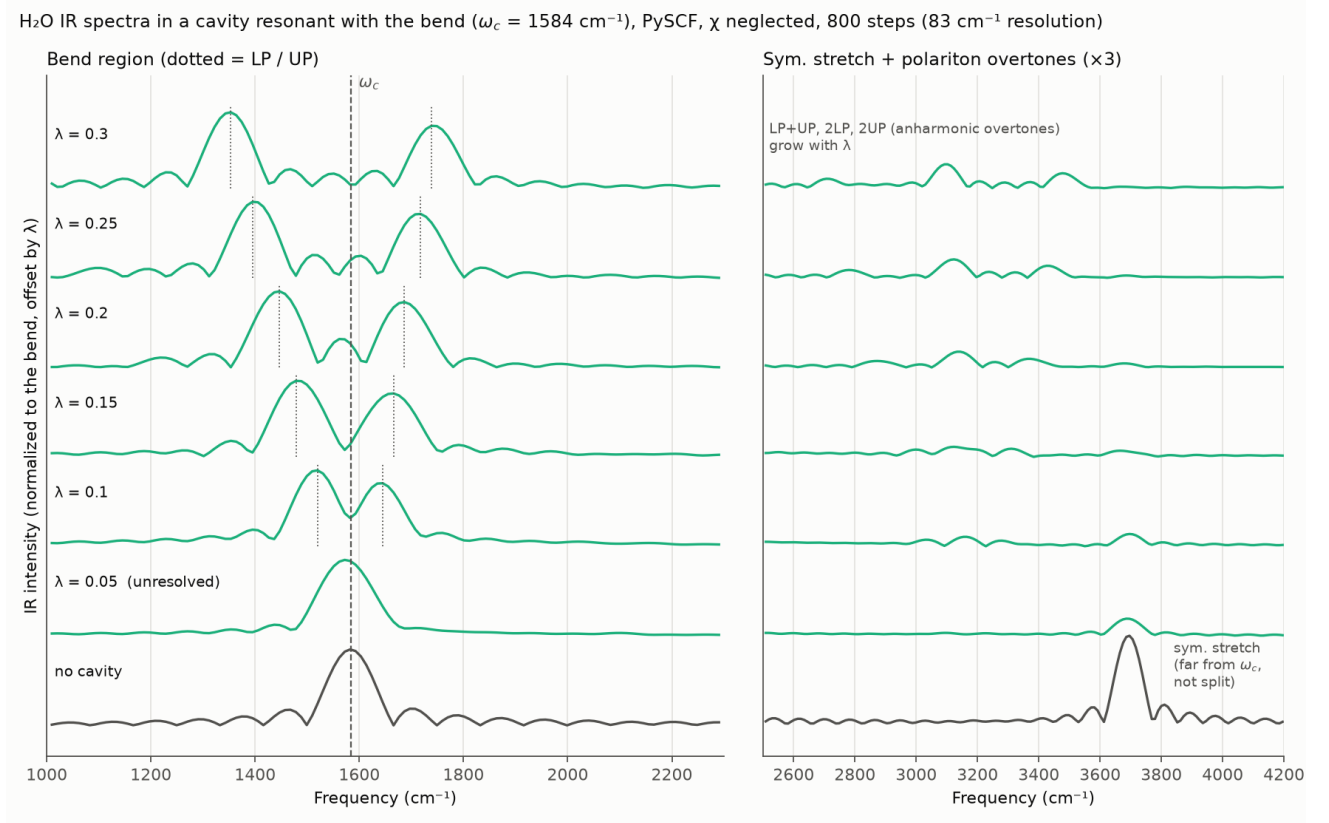


Figure 3: H₂O spectra, χ neglected, with the cavity resonant with the bend (1584 cm^{-1}). Left: the bend splits into LP and UP for $\lambda \geq 0.1$ (not resolved at $\lambda = 0.05$). Right: weak LP+UP, 2LP and 2UP overtones appear as λ grows. The symmetric stretch is far detuned and doesn't split.

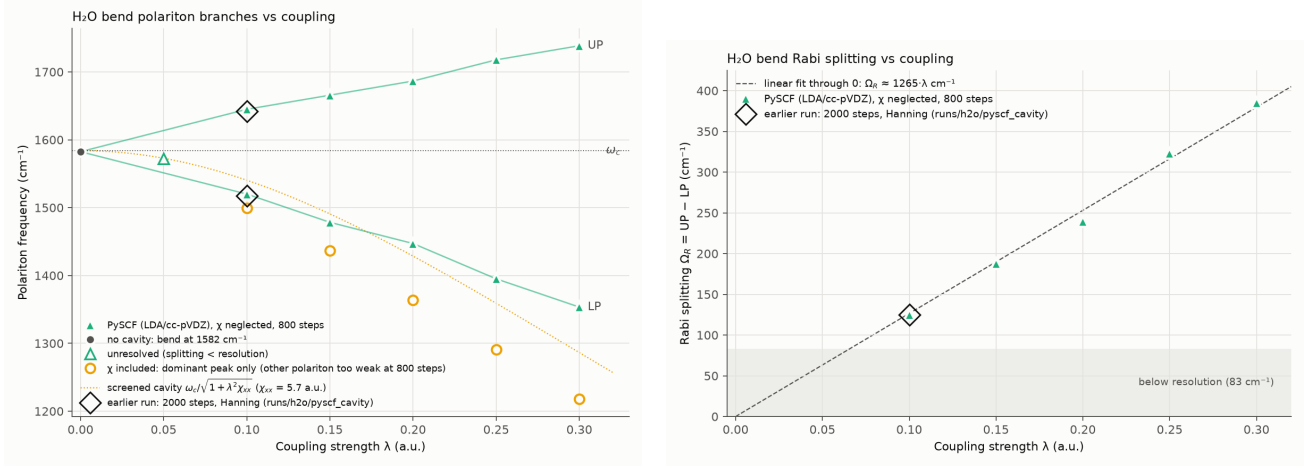


Figure 4: Left: H₂O branches. Green is χ neglected. Orange is χ included, dominant peak only, with the dotted line showing the screened cavity $\omega_c/\sqrt{1+\lambda^2\chi_{xx}}$. Right: Rabi splitting (χ neglected) with a linear fit. The diamond is the earlier 2000-step result.

λ	0.1	0.15	0.2	0.25	0.3
Ω_R , χ neglected (cm^{-1})	125	187	239	323	385
Dominant peak, χ included (cm^{-1})	1499	1437	1364	1291	1218
Screened cavity $\omega_c/\sqrt{1+\lambda^2\chi}$ (cm^{-1})	1541	1491	1429	1360	1286

Confirms: with χ neglected, Ω_R is linear in λ , and $\lambda = 0.1$ reproduces the earlier 2000-step result (125 cm^{-1}). With χ included ($\chi_{xx} \approx 5.7$ a.u.), the polarizability screens and red-detunes the cavity. One polariton dominates and tracks just below the screened cavity frequency, the same behaviour as the CO₂ χ -included benchmark. The second polariton is weaker than the finite-length side lobes at 800 steps, so there is no χ -on Ω_R yet.

3. Single-molecule checks

Energy conservation. All 67 runs conserve $H = E_{\text{kin}} + E_{\text{bare}} + \frac{1}{2}p^2 + \frac{1}{2}(1 + \lambda^2\chi) e_a^2$ to about 1% (drift $< 0.3\%$) at 0.5 fs, so no smaller timestep is needed. Note that `md.log`'s E_{tot} leaves out the cavity energy. Checked with `scripts/check_energy.py`.

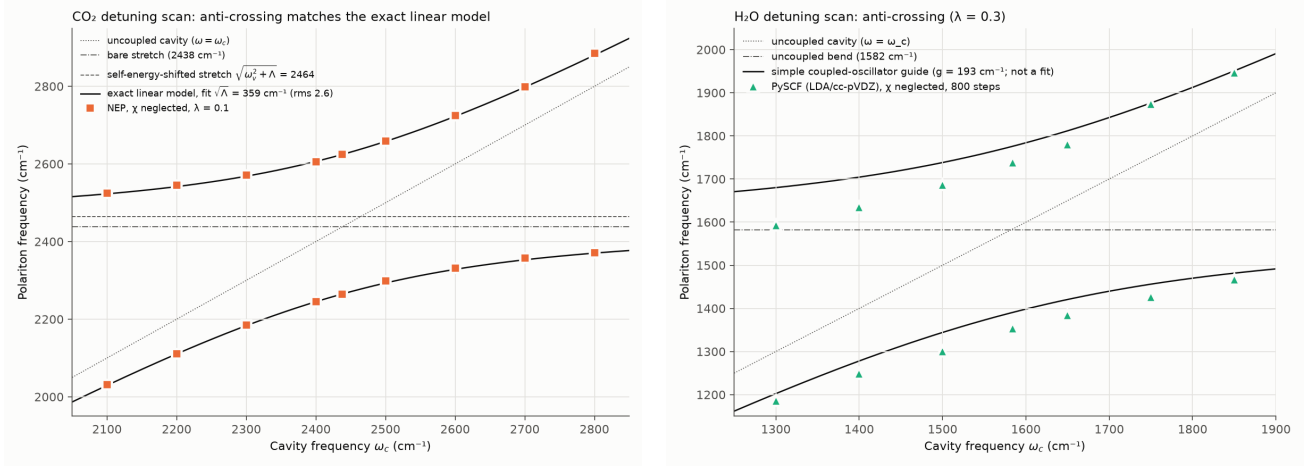


Figure 5: Detuning scans. Left: CO₂ (NEP, $\lambda = 0.1$). The exact linear model (vibration + cavity + dipole self-energy, one parameter) fits to 2.6 cm⁻¹ rms, and the effective stretch is shifted to 2464 cm⁻¹. Right: H₂O (PySCF, $\lambda = 0.3$). There is a clear anti-crossing, but the points sit 40–60 cm⁻¹ below the simple model.

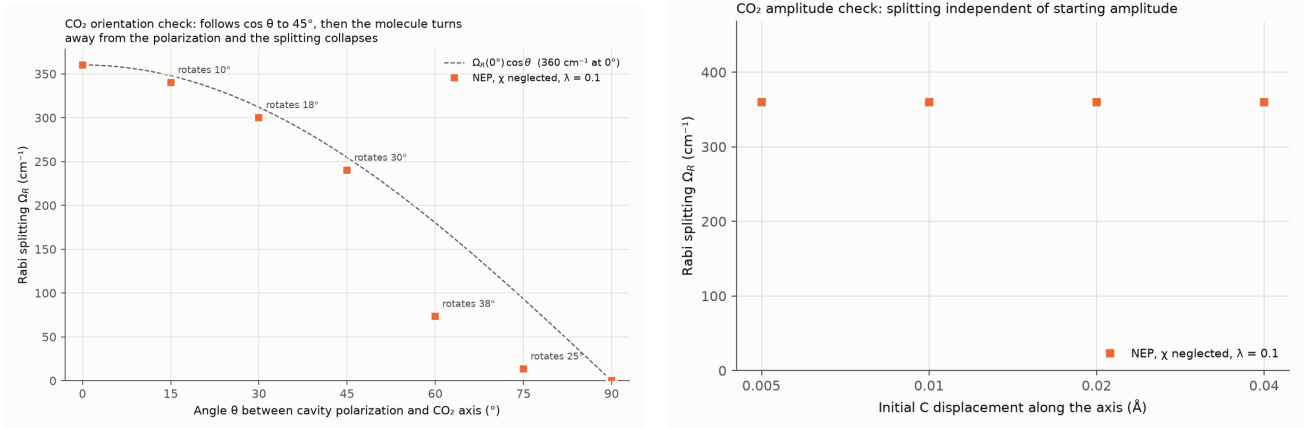


Figure 6: Left: CO₂ orientation. The polarization is tilted by θ from the molecular axis. Right: CO₂ splitting vs initial displacement.

Confirms:

- *Amplitude:* $\Omega_R = 360$ cm⁻¹ for C displacements of 0.005–0.04 Å, so the runs are linear and one trajectory per setting is enough.
- *Detuning:* CO₂ behaves exactly like the textbook linear model. True resonance is near 2500 cm⁻¹, not the bare 2438, because of the dipole self-energy.
- *Orientation:* CO₂ follows $\cos \theta$ to about 5% up to 45°. Beyond that, the molecule rotates away from the polarization (10–38°) and Ω_R collapses. H₂O, which has a permanent dipole, rotates even more (49° at $\theta = 30^\circ$, 113° at 60°), so it gave no clean $\cos \theta$ test.

Next steps

1. H₂O with χ on in the “resonant” variant ($\omega_c \rightarrow \omega_c \sqrt{1 + \lambda^2 \chi}$), or with longer runs plus a window, to resolve both polaritons.

2. Settle orientation and rotation (questions below) before choosing a 2–3 molecule geometry.
3. First multi-molecule test: 2–3 H₂O (PySCF), well separated ($> 10 \text{ \AA}$) and aligned with the cavity, to check $\Omega_R \propto \sqrt{N}$ against the single-molecule slope.

Questions

1. Is the molecular rotation in a tilted cavity field expected? Should orientation be fixed, or coupling kept lower, for multi-molecule runs, or is rotation part of the physics we want?
2. Should multi-molecule runs use χ with the resonant (un-screened) ω_c , as in `nep_chi_resonant`?
3. Which analytic model applies to the e-mode for a polar molecule? The H₂O detuning offset doesn't fit the linear dipole-self-energy model that works for CO₂.
4. Is the CO₂ NEP model valid for CO₂ pairs and clusters, or only single molecules?

Data, scripts and figures: github.com/charliew0707/VSC_Molecular_Dynamics. Campaign scripts are in `scripts/campaigns/`, plot scripts in `scripts/plot_*.py`, and peak tables in `results/**/peaks.csv`.